

51 Animal Behavior

KEY CONCEPTS

- 51.1** Discrete sensory inputs can stimulate both simple and complex behaviors p. 1140
- 51.2** Learning establishes specific links between experience and behavior p. 1143
- 51.3** Selection for individual survival and reproductive success can explain diverse behaviors p. 1148
- 51.4** Genetic analyses and the concept of inclusive fitness provide a basis for studying the evolution of behavior p. 1154

Study Tip

Identifying components of experimental design: Animal behavior is studied in both the wild and the laboratory and involves approaches ranging from passive observation to active intervention. As you study this chapter, note how scientific methods are applied to answer questions in diverse contexts, such as foraging in honeybees (Figure 51.5; see example started here) and fruit flies (Figure 51.13).

- 1) **Make observations**—returning bees do a number of different but set dances
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- 2) **Form a hypothesis**—the dances convey information about food location
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- 3) **Generate predictions**
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- 4) **Gather and analyze data**
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- 5) **Draw conclusions**

➔ Go to Mastering Biology

For Students (in eText and Study Area)

- Get Ready for Chapter 51
- HHMI Videos: Studying Elephant Communication; How Lizards Find Their Way Home

For Instructors to Assign (in Item Library)

- Interpret the Data: Evolution of Foraging Behavior
- Tutorial: Animal Behavior and Learning
- Scientific Skills Exercise: Crow Foraging Behavior



Figure 51.1 A male magnificent frigatebird (*Fregata magnificens*) lives mostly at sea. He often feeds by piracy, harassing diving birds into coughing up their catch for his consumption. Periodically, however, he alights on an island, points his bill to the sky, and inflates an enormous red throat pouch. Clacking his beak, he produces a drumming sound that resonates in the pouch. If a curious female takes notice and lands nearby, a mating ritual ensues.

What questions do biologists seek to answer in studying animal behavior?

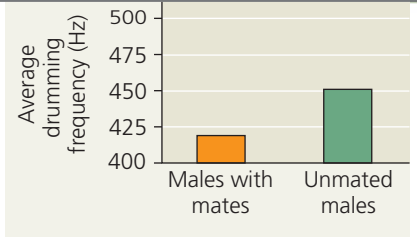
1. What event or signal triggers the behavior?

Males inflate their pouch only if females are nearby. The intensity of the display increases when a female circles overhead.



2. How does experience during growth and development influence the behavior?

Older males have larger pouches that produce lower drumming frequencies, which attract more females.



3. How does the behavior aid survival and reproduction?

After mating, males and females cooperate in nest building and feeding the chicks.



4. How has the behavior been shaped by natural selection?

The throat pouch can be inflated prominently for mating, and deflated to maximize flight efficiency.



Answering these questions can provide fundamental insights into **how** behaviors occur and **why** they arise.